CONDITION & CONSTRAINT

Structure Without Foundation

Anthony Vito Coppa

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Abstract

This paper isolates a minimal grammar of condition and constraint before truth, interpretation, quantity, or ontology is assigned. A condition declares an office in which a candidate can be addressed; a constraint marks the boundary within which that address can be posed again without collapse. The resulting cut is deliberately spare: admission is local to a declared contract, non-vacuity requires at least one retained distinction, and no chart may recover a distinction it has already erased. An inscriptional witness shows how recognizability can survive reuse without being identified with exact coincidence. A final conditional example records the familiar whole–remainder fixed point and its discriminant $\sqrt{5}$, while making explicit that this coordinate follows only after a self-similar chart has been chosen. The construction supplies no universal foundation. It gives a discipline for stating where a rule applies, what it preserves, and exactly where it stops.

1 Position

What follows is a constraint walk. Each step removes a form of statement that cannot survive its own declared use. The paper does not ask what exists. It asks the prior and smaller question: under what declared conditions can a candidate be posed, retained, and posed again without the rule of admission silently changing?

This order matters. Existence, truth, reference, measurement, and meaning may all become legitimate downstream offices. None is required to draw the first boundary. The present grammar therefore begins neither from a thing nor from a theory of things. It begins from a candidate address and a rule governing whether that address remains available.

2 Condition and constraint

DEFINITION 2.1 (Condition). A *condition* is a declared office of address

$$\mathcal{D} = (\Sigma, \mathcal{R}, \rho),$$

where Σ is a class of candidate determinations, $\mathcal{R}$ specifies the allowed occasions of re-posing, and ρ states the resolution at which retained distinctions are to be read. No internal structure on Σ is presumed.

DEFINITION 2.2 (Constraint). A *constraint* is a declared restriction on continuation in $\mathcal{D}$. Equivalently, it specifies an admissible subclass $A \subseteq \Sigma$, together with whatever evidence the declared office requires for membership in A.

The condition fixes where a term may lawfully apply. The constraint fixes what must hold there. Neither one supplies content merely by being named. Changing the domain, the re-posing rule, or the promised resolution creates a new version of the office; it does not retroactively alter the old one.

DEFINITION 2.3 (Admissibility). For $\sigma \in \Sigma$, write

$$\mathsf{can}_{\mathcal{D}}(\sigma)$$

when σ satisfies the declared constraint and remains available under the permitted re-posing operation. When continuation fails, write $\perp$ at the metalevel. The mark $\perp$ is not an additional element of Σ.

Thus the first cut is binary only after the office has been fixed: continuation is admitted or it is not. This is not a claim that the world is binary, nor a valuation of the candidate. It is the yes–no output of one declared gate.

3 Re-posing without imported time

Let $R(\sigma, \tau)$ mean that τ is an allowed re-posing of σ under $\mathcal{R}$. The relation R need not be temporal, causal, total, or deterministic. It may describe another inscription, another coordinate presentation, another run of a procedure, or another lawful use.

Let $\sigma \sim_\rho \tau$ mean that the two remain recognizable as belonging to the same declared class at resolution ρ. This is not exact equality. It is not reference to an object outside the office. It is the minimum relation needed to say that re-posing has not become indistinguishable from replacement.

The non-vacuity floor is then explicit.

DEFINITION 3.1 (Non-vacuity). The office $\mathcal{D}$ is non-vacuous at ρ when there exist $\sigma, \tau \in \Sigma$ such that

$$\mathsf{can}_{\mathcal{D}}(\sigma), \qquad \mathsf{can}_{\mathcal{D}}(\tau), \qquad R(\sigma, \tau), \qquad \sigma \not\sim_\rho \tau,$$

while both addresses remain legible in the office.

This requirement rules out the one-class quotient as a free victory. A constant gate can certify nothing about a distinction that the office promises to preserve. The floor is fixed by the promise, not loosened after the read.

4 An inscriptional witness

The abstract cut can be witnessed without yet introducing number or measure. Let M be a class of legible marks and let $\oplus$ be a partial operation of re-inscription. The signs below are used

only in their ordinary formal roles:

$$\begin{aligned} \equiv &\quad \text{exact coincidence,} \\ \neq &\quad \text{non-coincidence,} \\ \sim_\rho &\quad \text{recognition at the declared resolution.} \end{aligned}$$

For a mark $x \in M$, two limiting cases are immediately visible. Exact closure has the form

$$x \oplus x \equiv x.$$

Re-inscription has left no legible remainder at the declared resolution. Nontrivial extension has the form

$$x \oplus x \neq x, \qquad x \sim_\rho (x \oplus x).$$

The first relation retains difference; the second retains continuity of read. Together they witness extension without replacement.

PROPOSITION 4.1 (Witness triad). *In this inscriptional chart, the three roles*

$$x \equiv x, \qquad x \oplus x \neq x, \qquad x \sim_\rho (x \oplus x)$$

are jointly sufficient to display coincidence, nontrivial reuse, and recognizability without coincidence.

Proof. The first expression supplies a stable coincidence test. The second supplies a pair distinguished at the declared resolution. The third prevents that pair from being read merely as unrelated replacement. No stronger classification is inferred. □

The proposition is a witness, not an exhaustive ontology of persistence. It introduces no arithmetic law for $\oplus$, no identity element, and no claim that every admissible office has this presentation.

5 The chart cannot restore what it erases

A chart is not innocent merely because it is readable. Once a chart is used as the public carrier of an audit, its information loss bounds every later reader.

Let H be a history class, let $\sim_\rho$ be the equivalence relation on H induced by the declared resolution, let $\pi_\rho : H \to H/\!\sim_\rho$ be its quotient map, and let $W : H \to \Sigma_W$ be a chart or writer. A reader is any function $C : \Sigma_W \to H/\!\sim_\rho$.

LEMMA 5.1 (Erasure). *If $h_1 \not\sim_\rho h_2$ but $W(h_1) = W(h_2)$, then no reader C can satisfy*

$$C \circ W = \pi_\rho$$

on both h_1 and h_2.

Proof. Because $W(h_1) = W(h_2)$, functionality gives $C(W(h_1)) = C(W(h_2))$. But $h_1 \not\sim_\rho h_2$ gives $\pi_\rho(h_1) \neq \pi_\rho(h_2)$. The two equalities are incompatible. □

The lemma gives the exact obstruction. If a promised difference is collapsed into one mark, later interpretation cannot recover it. Conversely, preserving every raw difference is not automatically useful: the office must still declare which differences matter and how they are read. An admissible chart therefore needs both a non-degenerate resolution floor and a legible rule of return.

6 Structure without foundation

The word *structure* names the relations that remain once a condition and its constraints have been declared. It does not name an absolute substrate beneath every possible office. In particular, this paper supplies no view from outside all conditions by which every condition could be ranked.

That absence is not a defect to be repaired by a hidden foundation. It is a versioning rule. Every assertion carries its office:

$$(\Sigma, \mathcal{R}, \rho, A;\ \text{rule, evidence, result}).$$

Two results may be compared only after a typed map between their offices has been given. Repeated vocabulary is not such a map.

The consequence is conservative. A result may be exact inside its declared condition without becoming universal. A failed condition may be refused without converting that refusal into a claim about everything outside its domain.

7 A conditional whole–remainder chart

The original composition associated interior traversal with $\sqrt{5}$. That association is valid only after an additional chart contract has been supplied. The minimum exact form is as follows.

Assume a positive coordinate r is required to reproduce its own whole–remainder decomposition:

$$r = 1 + \frac{1}{r}.$$

This is a self-similar two-part chart. It is not a consequence of admissibility alone.

PROPOSITION 7.1 (Whole–remainder fixed point). *Under the stated chart contract, the unique positive fixed point is*

$$r = \frac{1 + \sqrt{5}}{2}.$$

Proof. Multiplication by $r > 0$ gives $r^2 - r - 1 = 0$. The quadratic formula gives $r = (1 \pm \sqrt{5})/2$, of which exactly one root is positive. □

Here $\sqrt{5}$ is the discriminant of a particular fixed-point equation. It does not follow from the bare admissibility cut, and it is not the unique coordinate of every non-closing traversal. The proposition preserves the early chart as a lawful witness while keeping its additional assumption visible.

8 Exact boundary

The construction establishes the following.

i. Admission is meaningful only relative to a declared condition.
ii. A non-vacuous office must preserve at least one distinction fixed by the claim it promises to certify.
iii. Reuse may retain recognizability without requiring exact coincidence.

iv. No downstream reader can reconstruct a promised distinction erased by the chart that carries it.
v. Numerical or geometric witnesses enter only through separately stated chart contracts.

The paper does not establish a theory of truth, a semantics, an ontology, a dynamics, or a universal geometry. It does not derive π, Φ, Euler's formula, or physical law from admission. Those objects can enter later only through their own definitions and typed adapters.

The stopping point is therefore exact: a rule must say where it applies, what distinction it preserves, and what would count as failure. A reference may then orient a later use. Neither the rule nor the reference may supply the missing content by implication.

Methodological orientation

Wittgenstein's rule-following discussions and his discipline concerning the limits of expression are relevant orientation for this method, especially the warning that a rule cannot secure its own application by an indefinitely added interpretation. They are not premises of the formal lemmas above. The present paper's narrower device is a typed refusal: when the domain, resolution, or recovery map is absent, the corresponding claim is not yet formed.

Provenance

The proof composition is dated 22 January 2026; a surviving early typeset copy carries the date 23 January 2026. This canonical reading edition was prepared on 22 August 2026. It preserves the original admissibility cut, the inscriptional witness, the charting concern, and the title *Structure Without Foundation*; it narrows claims of exhaustion and makes the whole–remainder chart explicitly conditional. Later results in the wider corpus do not function as premises for the January argument.

References

[1] Ludwig Wittgenstein, *Philosophical Investigations*, 4th ed., trans. G. E. M. Anscombe, P. M. S. Hacker, and Joachim Schulte, Wiley–Blackwell, 2009, esp. §§198–202.

[2] Ludwig Wittgenstein, *Tractatus Logico-Philosophicus*, trans. D. F. Pears and B. F. McGuinness, Routledge, 1961, esp. propositions 6.5–7.